Mars Habitat Robots!

Your engineering team has been tasked with the design of robots to help build a human habitat on mars.

You must build an autonomous robot that can:

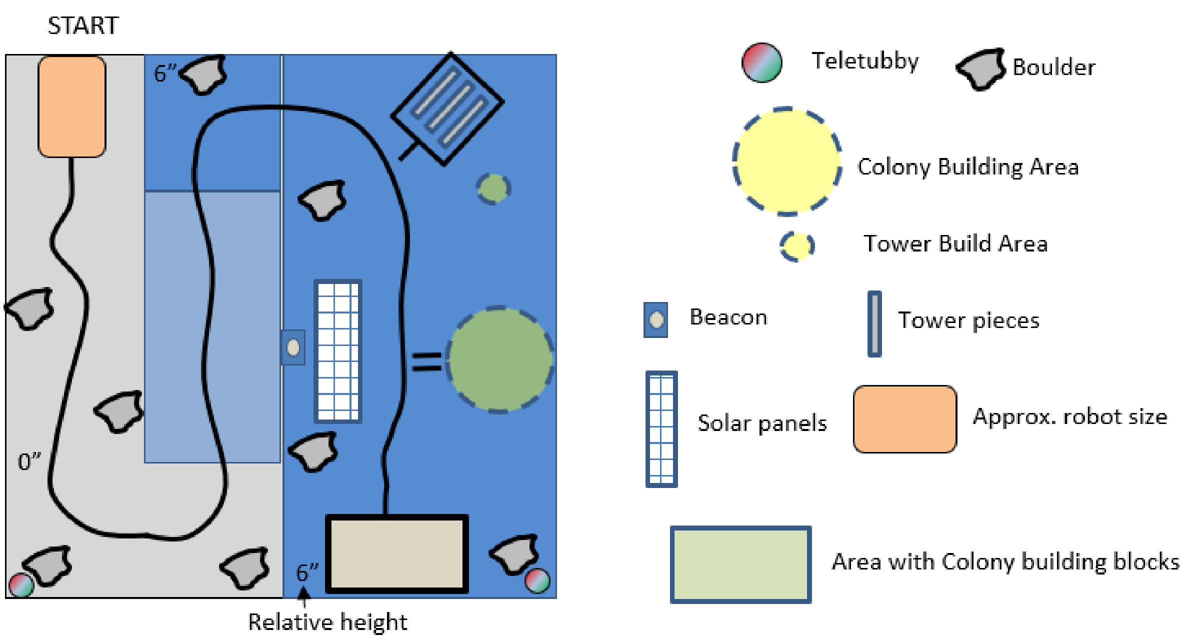
- Cross rough terrain to get to the habitat site
- Build the habitat by assembling pre-made modules
- Build a radio tower by assembling nesting mast pieces
- Activate solar panels by removing their protective cover
- Search for and collect minerals for future manufacturing

Do you have that weird feeling like someone's watching you? The Teletubbies have returned and are hiding on Mars. If you see one, sound the alarm!

Revision History:

1.0 AM initial draft

Competition Surface (Approximate! Will evolve during construction)



COMPETITION RULES

1) **Play Surface** - The play surface is approximately 8 feet x 8 feet, containing a ramp and an elevated floor that is 6” above the surface.

The surface is made of wood and will have some warp and slight bumps at the joints and at the ramps. Robots must be designed to accommodate for imperfections and irregularities in the surface.

There are two Play Surfaces, one for each robot, in order to compete head-to-head during each heat. Although the surfaces will follow the same guidelines and landmarks, there is no guarantee that the surfaces will be made to match identically to each other.

2) **Path** – There is a black electrical tape path as shown. The path will have at least 6” clearance to obstacles.

3) **Start Area** - The robots will start at the marked end of each tape track.

4) **<u>Boulders and minerals</u>** – Large (plastic) rocks of various dimensions will be located on the surface in random locations. One of these rocks will contain a ferrous metal which can be detected electromagnetically. The rock containing the metal will not be visibly different from the others. Robots must identify this rock, and collect it onto the robot for points.

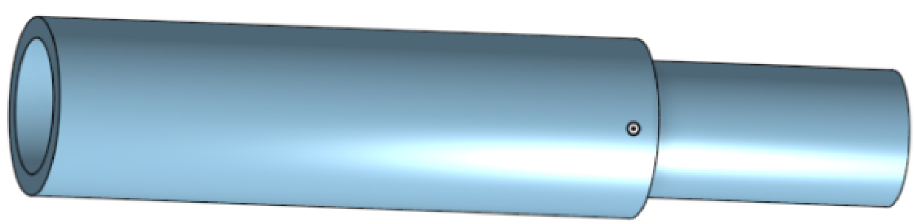
5) **<u>Colony Building Area</u>** - An approximately 12" circular area will be marked with a dashed electrical tape perimeter and with two 3" tape pieces located 1" from the tape track and 1" apart.

6) **<u>Tower Building Area</u>** - An approximately 4" circular area will be marked with a dashed electrical tape perimeter and will be located 2" to the right of the Tower Piece Area

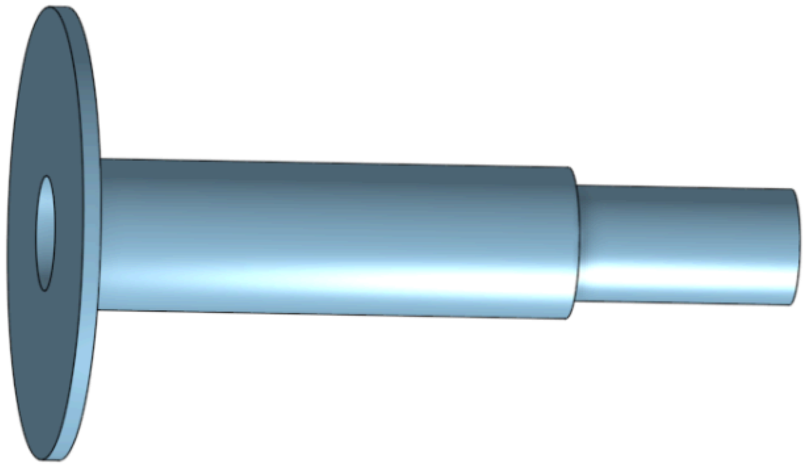
7) **<u>Tower Piece Storage Area</u>** - An approximately 6" square area will be marked with an electrical tape perimeter and with one 3" tape pieces located 1" from the tape track.

8) **<u>Tower Pieces</u>** – 1" Diameter, 5" long tubes meant to stack by having a thin end that can insert into a wider end. STL files will be provided for 3D printing.

Two of these:

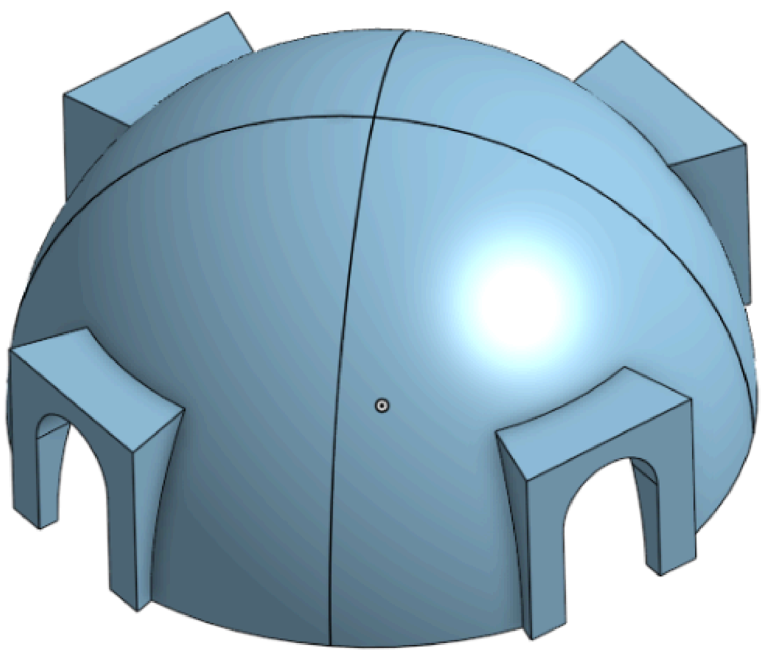


One of these:



9) **Habitat Piece Storage Area** - An approximately 6” x 12” area will be marked with an electrical tape perimeter and will be located at the end of the tape track.

10) **Habitat Pieces** – Four identical pieces that together make a dome structure. Magnets on both sides hold them together when assembled. STL files will be provided for 3D printing.



11) **Solar Panels** – Solar panels measuring approximately 3” x 10” are covered with a plastic cover that is held in place by Velcro. This cover has three loops approximately

1” diameter to assist with removal. The location of the panel is only determined by the beacon.

12) **Panel beacon** – An Infrared beacon emitting ~1um infrared light pulses at 1 kHz will be located behind the panel as shown.

13) **Teletubbies**- Two Teletubby or similar dolls will be hiding behind some of the boulders. To “Discover” them, your robot must be obviously pointing toward them and a light on the robot must flash at least three times. If this light flashes at another time when no teletubbies are in the robot’s direction (false positive), all points for teletubbies discovered are nulled.

14) **Scoring** – Scores after 2 minutes are calculated as follows:
    a) 1 point for each habitat piece assembled
    b) 1 point for each tower piece assembled
    c) 2 points for uncovering the solar panels
    d) 1 point for each Teletubby discovered (0 for any false alarm)
    e) 2 points for recovering iron (0 if non-ferrous rocks are recovered)

15) **Restarting Robots** - Robots may be rescued by the team and restarted as many times as desired during the heat, for any reason. Components must be reset after a restart. The maximum points reached for any attempt during the heat will count as the score.

16) **Time Limit** – Heats are a maximum of 2 minutes. Additional time may be allowed in the finals at the judges’ discretion to resolve close heats. Judges may choose to end a heat early if there is a clear winner during the heat or robots are not performing.

General Rules

A) **Autonomy:** Robots must be completely autonomous – no form of remote control is allowed.

B) **Size:** The Robot must be small enough to pass through the boulders and may not extend at any time beyond 18” x 18” x 18” (TBD)

C) **Power:** Robots may only be powered by batteries and stored elastic energy.

D) **Components:** All components outside of those provided by the course instructors or listed at the end of this document must be approved by course instructors. Teams that choose to purchase their own items will not be reimbursed and are limited to a maximum of $200 per team.

E) **Damage to the Surface:** Robots may not permanently modify or damage the competition surface or any individual competition piece including stuffed animals. No glues, nails, screws or similar "rough measures" are allowed to pick up the pets.

F) **Start Mechanism:** Robots will initiate motion only when the START button on their controller is pressed by a team member at the start of the run (signalled by a Judge).

G) **Competition Surface Variations:** The surface is made of wood and will have some warp and slight bumps all over. Robots must be designed to accommodate for imperfections and irregularities, as well as variation between practice and final surfaces. The surface will be painted at the beginning of the course but will not be repainted before competition day to avoid sudden changes in reflectivity. The surface will be fragile. **NO STANDING OR SITTING ON THE SURFACE!** Don't use tools or work on your robot while on the surface. Design your robot in such a way that it won't damage any parts of the surface while navigating it.

H) **Rules Finalization:** Rules and dimensions may change between now and the competition. Finalized rules will be issued as early as possible. Qualifying heats ("Time Trials", with no opponent) will take place 2 weeks prior to the competition on a final version of the surface.

I) **Sportsmanship Rules:** Strategies or designs that obviate the design elements of the course or that do not follow the intent of the competition will be disallowed whether or not they explicitly break these rules. All strategies which have been designed specifically to come as "close to" violating any of the posted rules as possible must be presented to the course instructors during the design stage of robot building. All decisions are at the discretion of the course instructors.

ALLOWED AND RESTRICTED MATERIALS

Approved

1. Solenoids (when used with mechanical constraints).
2. Elastic bands.
3. Wheels and hubs from existing RC or other small vehicles.

Must be Reviewed By Course Instructors

1. Springs are generally allowed, but must be reviewed individually for safety.
2. Compressed air may be allowed, but all valves and fittings must be reviewed for safety and a maximum pressure limit will be imposed.
3. Usage of existing CAD models. If not reviewed and approved by an instructor, usage of these will be considered plagiarism.

Banned:

1. Discrete H-bridge driver chips.
2. Any components other than wheels and hubs from existing RC or other small vehicle chassis, including (but not limited to) suspensions, differentials, steering mechanisms.